

1. write a program to print all even numbers from 1 to 50 using a for loop in a single line.

```
for (let i=1; i<=50; i++) {  
  if (i%2===0) {  
    console.log(i);  
  }  
}
```

2. Factorial of a number using while loop

```
const factorial = (n) => {  
  let result = 1;  
  let i = 1;  
  while (i <= n) {  
    result *= i;  
    i++;  
  }  
  return result;  
};
```

```
console.log(factorial(5));
```

3. print numbers from 20 to 1 using do..while loop

```
let i = 20;  
do {  
  console.log(i);  
  i--;  
} while (i >= 1);
```


4. Reverse the string "JavaScript" without using reverse()

```
let str = "JavaScript";  
let reversed = "";  
for (let i = str.length - 1; i >= 0; i--) {  
    reversed += str[i];  
}  
console.log(reversed);
```

5. Find the highest number in an array using a loop.

```
const numbers = [45, 12, 89, 34, 67, 90, 23];  
let max = numbers[0];  
for (let i = 1; i < numbers.length; i++) {  
    if (numbers[i] > max) {  
        max = numbers[i];  
    }  
}  
console.log("Highest number: ", max);
```

6. print only odd numbers and find their total

```
const numbers = [10, 15, 20, 25, 30, 35, 40];  
let total = 0;  
for (let i = 0; i < numbers.length; i++) {  
    if (numbers[i] % 2 !== 0) {  
        console.log(numbers[i]);  
        total += numbers[i];  
    }  
}  
console.log("Total of odd numbers: ", total);
```


7. Create a student object with name, age, course, mark.

```
const student = {  
  name: "Rithika",  
  age: 22,  
  course: "Full stack Development",  
  mark: 88  
};
```

```
console.log(student.name);  
console.log(student.age);  
console.log(student.course);  
console.log(student.mark);
```

```
console.log(student["name"]);  
console.log(student["mark"]);
```

8. Display students who scored more than 75

```
const students = [  
  { name: "Aren", mark: 85 },  
  { name: "Bala", mark: 65 },  
  { name: "Rithika", mark: 90 },  
  { name: "Gobika", mark: 70 }  
];
```

```
for (let i = 0; i < students.length; i++) {  
  if (students[i].mark > 75) {  
    console.log(students[i].name, "-",  
                 students[i].mark);  
  }  
}
```


9. Square of a number: Normal function, function Expression, Arrow Function.

① Normal Function

```
function squareNormal (num){  
  return num * num;  
}
```

② Function expression

```
const squareExpr = function (num){  
  return num * num;  
};
```

③ Arrow function

```
const squareArrow = (num) => num * num;  
console.log (squareNormal (5));  
console.log (squareExpr (5));  
console.log (squareArrow (5));
```

10. Function with name & age → formatted sentence using template literal.

```
function greet (name, age){  
  return 'My name is ${name} and  
  I am ${age} years old.';  
}  
console.log (greet ("Ravi", 25));
```

11. var, let, const inside the block scope

```
{  
  var x = 10;  
  let y = 20;  
  const z = 30;  
  console.log (x, y, z);  
}  
console.log (x);  
console.log (y);  
console.log (z); // Error.
```


12. predict the output

```
console.log(a);  
var a=10;
```

```
console.log(a);
```

```
var a=10;
```

o/p: undefined - because var is hoisted.

```
console.log(a);
```

```
let a=10;
```

o/p: Reference Error.

13. Destructuring: extract name and course.

```
const student = {
```

```
  name: "Ravi",
```

```
  age: 22,
```

```
  course: "JavaScript"
```

```
};
```

```
const {name, course} = student;
```

```
console.log(name);
```

```
console.log(course);
```

14. Arrow function with rest parameter - Sum of multiple numbers.

```
const sumAll = (...numbers) => {
```

```
  let total = 0;
```

```
  for (let num of numbers) {
```

```
    total += num;
```

```
  }
```

```
  return total;
```

```
};
```

```
console.log(sumAll(10, 20, 30, 40));
```

15. Users : display name, city, total marks

```
const users = [
```

```
{
```

```
  name : "Ravi",
```

```
  marks : [80, 90, 85],
```

```
  address : {
```

```
    city : "chennai"
```

```
  }
```

```
},
```

```
{
```

```
  name : "kumar",
```

```
  marks : [60, 70, 85]
```

```
}
```

```
];
```

```
users.forEach(( { name, marks, address } ) => {  
  const city = address?.city ?? "city Not  
    Available";
```

```
  const total = marks.reduce((sum, mark) =>  
    sum + mark, 0);
```

```
  console.log(`${name} - ${city} - ${total}`);  
});
```